## Ionisation enthalpy

Alkali metals have the lowest ionisation enthalpy compared to other elements present in the respective period. As we go down the group, the ionisation enthalpy decreases due to the increase in atomic size. In addition, the number of inner shells also increases, which in turn increases the magnitude of screening effect and consequently, the ionisation enthalpy decreases down the group.

The second ionisation enthalpies of alkali metals are very high. The removal of an electron from the alkali metals gives monovalent cations having stable electronic configurations similar to the noble gas. Therefore, it becomes very difficult to remove the second electron from the stable configurations already attained.

## Hydration enthalpy

Lithium salts are more soluble than the salts of other metals of group 1. eg. $LiClO_4$ is upto 12 times more soluble than $NaClO_4$. Other salts $KClO_4$, $RbClO_4$ and $CsClO_4$ have solubilities only $10^{-3}$ times of that of $LiClO_4$ . The high solubility of Li salts is due to strong solvation of small size $Li^+$ ion.

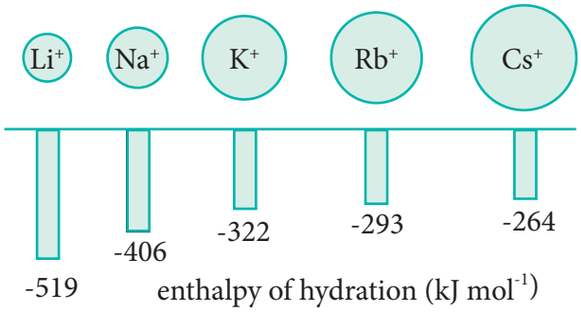



**Figure 5.2 Hydration enthalpy of alkali metals**

## Electronegativity:

Alkali metals have comparatively smaller value of electronegativity than the other elements in the respective period. When they react with other elements, they usually produce ionic compounds. For example, they react with halogens to form ionic halides.

## Flame colour and the spectra:

When the alkali metal salts moistened with concentrated hydrochloric acid are heated on a platinum wire in a flame, they show characteristic coloured flame as shown below.

**Table 5.4 Flame colour and wavelength**

| Element | Colour | Wavelength (nm) |
|---|---|---|
| Lithium | Crimson red | 670.8 |
| Sodium | Yellow | 589.2 |
| Potassium | Lilac (violet) | 766.5 |
| Rubidium | Reddish violet | 780.0 |
| Caesium | Blue | 455.5 |

The heat in the flame excites the valence electron to a higher energy level. When it drops back to its actual energy level, the excess energy is emitted as light, whose wavelength is in the visible region as shown in the above table.

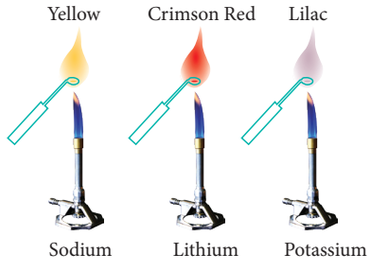



**Figure 5.3 Flame colours of alkali metal salts**



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